

Paper_ED_Dyn_01 — Saddle Dynamics: Foundational Framework

Series: Wave-3 Generative Papers (Dynamics Arc — foundational paper) **Status:** Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** within DCGT A→regime. Direct composition of Paper_ED_SC_4_9 Hessian framework + Route A4 substrate-parameter inheritance (Paper_027-analog); no new primitives or paper-specific postulates. **Date:** 2026-05-17 **Authors:** Paper_087 (13 primitives); Paper_073 (DCGT); Paper_ED_SC_4_9 (S1/S2/S3 classification); Paper_089/090/093 (V1/V5/T18 kernels); Paper_012 (substrate-c); Paper_027 (Newton’s G template); Paper_ED_Cos_01 (Inflation, M3); Paper_ED_Cos_05 (Dark Energy, M3); Paper_ED_Dyn_02 (Horizon-Motion, M2); Paper_ED_Dyn_03 (Radiation Law, M2); Paper_ED_GW_00 (GW, M3); Paper_095 (verdict grammar). Substrate-research support memos (Q1Q2 + Chain-Class + Route A project) cross-referenced in §2.

Label-class note (per Paper_095 §2.2, pending formal addition): This paper uses two sub-labels of **I** (inherited) introduced across the 2026-05 corpus push and applied across Cos_05, Cos_06, and this paper: **I (substrate-parameter-INHERITED)** = inheritance of a numerical value through substrate-parameter content (e.g., $\Theta_{\text{ED}}^{\text{Planck-units}}$, τ_{V_5}/τ_P) at the Paper_027-analog level; **IC-INHERITED** = inheritance of initial-condition / boundary-state content from an upstream substrate-state specification (framework-neutral per Memo_ED_CCC_Realizability_SubConstruct). A formal Paper_095 §2.2 update is the preferred corpus-wide grounding; the present in-paper note is the bridging definition.

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of saddle classification or Hessian-signature partition from nothing. The S1/S2/S3 classification is inherited from Paper_ED_SC_4_9; this paper applies that classification to *dynamics* specifically — i.e., the substrate-graph response under V1 + V5 propagation around saddle configurations.
2. It does **not** sit within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). ED is in the substrate-ontology research program (’t Hooft cellular-automaton; Wolfram Ruliad; causal-set program, Sorkin et al.). Methodology per Paper_095 (form-FORCED / value-INHERITED).
3. It does **not** introduce new substrate primitives or paper-specific postulates. The dynamical content is composition of the SC-4.x Hessian framework + V1/V5 kernel propagation + Route A substrate-parameter-INHERITED scale, all of which are corpus-internal.
4. It does **not** claim *substrate-FORCED* exhaustiveness of saddle-dynamical regimes. The classification of dynamical regimes (compression-dominated collapse; expansion-dominated expansion; mixed-signature horizon dynamics; sign-flip oscillation) follows the Hessian-signature partition of SC-4.9 by composition; exhaustiveness in the substrate sense is inherited from SC-4.9’s OPEN status on that question.
5. It does **not** posit a fundamental metric or absolute time. Substrate has no fundamental metric (Paper_ED_GW_00 §3.2); substrate-time enters via commitment-rate $\hbar$ per Paper_012 P-RB-1. Saddle dynamics is *intrinsic to the substrate-action* $S_{\text{sub}}[\Psi]$, not to any pre-existing geometric background.
6. It does **not** address radiation-source-class identification (Route C1 / C4) or non-saturation

chain-class identification beyond what Paper_ED_Dyn_02 + Paper_ED_Dyn_03 cover. Dyn_01 supplies the *foundational saddle-dynamics framework* on which Dyn_02/Dyn_03/GW_00 build; load-bearing OPENs in those papers remain OPEN at the same level.

Abstract

The substrate-graph **saddle configuration** of the substrate-action $S_{\text{sub}}[\Psi]$ (Paper_ED_SC_4_9) is the **foundational generator of ED dynamical behavior** within the SC-4.9 S1/S2/S3 classification (substrate-FORCED exhaustiveness OPEN per Paper_ED_SC_4_9; see Preamble item 4). At each substrate locus, the Hessian eigenvalue-signature partitions orthogonal directions into compression-dominant ($\lambda_i < 0$, substrate-action concavity) and expansion-dominant ($\lambda_i > 0$, substrate-action convexity) axes; the saddle's signature-pattern at a locus, together with V1 + V5 kernel propagation, determines the local dynamical response. **Three regime classes** emerge by composition with the SC-4.9 S1/S2/S3 classification: (i) all-compression (S1-class) $\rightarrow$ collapsing / gravitational-collapse regime; (ii) all-expansion (S3-class) $\rightarrow$ expansion / inflationary or dark-energy regime; (iii) mixed-signature (S2-class) $\rightarrow$ horizon-class dynamics and radiation propagation. Form-IDENTIFIED via SC-4.9 Hessian framework + Q1A + Q2A + C3 substrate-graph chain (Memo_ED_Q1Q2_JointClosure_Construct). Value-INHERITED at substrate-parameter level: characteristic length $\ell_{\text{saddle}} \sim \ell_P(\Theta_{\text{ED}}^{\text{Planck-units}})^{-1/2}$ via Route A4 (Paper_027-analog substrate-parameter inheritance per Memo_ED_RouteA_MultiRouteConvergence_Audit option (ii)). IC-INHERITED from local Ψ -gradient structure per substrate state. **Verdict per Paper_095: M3.** Substrate-c bound preserved throughout. **F1 (sharpest):** observation of dynamical behavior at saddle locus inconsistent with the Hessian-signature partition (e.g., compression-dominant behavior at all-expansion S3-saddle) refutes the saddle-as-foundational-generator identification. Enables Paper_ED_Dyn_04 (gravitational collapse), Paper_ED_Dyn_05 (inspiral dynamics), Paper_ED_GW_01 (BH ringdown spectroscopy) as direct extensions.

§1 Statement of Result

Substrate-graph dynamics at any locus is generated by the substrate-action saddle's Hessian-signature structure at that locus, composed with V1 retarded propagation (Paper_089) + V5 cross-chain coupling (Paper_090) carrying the local response outward at substrate-c.

Three regime classes by Hessian signature (exhaustive within SC-4.9 S1/S2/S3, not substrate-FORCED-exhaustive):

Hessian signature	SC-4.9 class	Dynamical regime
All eigenvalues negative ($\lambda_i < 0$ all i)	S1 (all-compression)	Collapsing / gravitational-collapse dynamics (Paper_ED_Dyn_04 territory)

Hessian signature	SC-4.9 class	Dynamical regime
All eigenvalues positive ($\lambda_i > 0$ all i)	S3 (all-expansion)	Expansion dynamics — inflationary (Cos_01) and late-LDE dark-energy (Cos_05) regimes
Mixed signature (some $\lambda_i < 0$, some $\lambda_i > 0$)	S2 (saddle proper)	Horizon-class dynamics (Dyn_02) and propagating signature-reconfiguration (GW_00)

FORM: saddle-Hessian-signature-driven dynamics — **IDENTIFIED** via SC-4.9 Hessian framework + Q1A + Q2A + C3 substrate-graph chain. Zero pure-D rows; composition + identification.

VALUE: characteristic length $\ell_{\text{saddle}} \sim \ell_P(\Theta_{\text{ED}}^{\text{Planck-units}})^{-1/2}$ **INHERITED at substrate-parameter level** via Route A4 (Paper_027-analog; per Memo_ED_RouteA_MultiRouteConvergence_Audit option (ii)). At saturation regime (uniform Ψ), $\ell_{\text{saddle}} \rightarrow R_H = c/H_0$.

IC: local Hessian-signature pattern + local Ψ -gradient structure **INHERITED** per substrate state (framework-neutral; cyclic-CCC vs single-aeon outside scope per Memo_ED_CCC_Realizability_SubConstruct).

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime per §4 audit table.

§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02 (participation), P03 (translation invariance — Noether source), P09 (polarity), P10 (multiple rule-types — V1, V5 kernel structure), P11 (commitment / time arrow), P12 (ED-threshold + stability landscape — Hessian-signature relevance), P13 (translation arrow).

Upstream papers: - Paper_012 (P-RB-1 substrate-c bound) - Paper_027 (Newton’s G via dimensional rearrangement — substrate-parameter inheritance template) - Paper_073 (DCGT hydrodynamic-window coarse-graining) - Paper_089 (V1 retarded kernel + T18 forward-causal) - Paper_090 (V5 cross-chain finite-memory) - Paper_093 (T18 kernel-arrow direction) - Paper_ED_SC_4_9 (saddle-geometry classification S1/S2/S3; Hessian-signature partition + Morse-theory machinery) - Paper_ED_Cos_01 + Paper_ED_Cos_05 (saturation-regime substrate-graph chain Q1A + Q2A + C3 + IC-INHERITED) - Paper_ED_Dyn_02 + Paper_ED_Dyn_03 (non-saturation regime + time-varying source extensions; M2 pending Routes C1 + C4) - Paper_ED_GW_00 (saddle-Hessian propagation framework for GW phenomenology) - Update_ED_SC_4x_Arc_M3 (six-projection arc-wide M3 upgrade post-Route-A) - Standard cosmology / standard physics (textbook inheritance where invoked)

No new substrate primitives. No paper-specific postulates. The dynamical content is composition of corpus-internal upstream content.

§3 Derivation

§3.1 Hessian sign structure at substrate locus

Per Paper_ED_SC_4_9, the substrate-action Hessian $\mathcal{H}(\ell, \ell') = \delta^2 S_{\text{sub}} / \delta \Psi(\ell) \delta \Psi(\ell')$ at each substrate locus has eigenvalue decomposition

$$\mathcal{H} \vec{u}_i = \lambda_i \vec{u}_i, \quad i = 1, \dots, d_{\text{sub}}$$

where $\vec{u}_i$ are eigenvectors in the substrate-graph direction space and λ_i are eigenvalues. The eigenvalue-signature pattern $\text{sign}(\lambda_1, \dots, \lambda_d)$ partitions substrate directions into compression-dominant ($\lambda_i < 0$, substrate-action concavity — local *contraction* under V1/V5 dynamics) and expansion-dominant ($\lambda_i > 0$, convexity — local *expansion*).

This Hessian-signature classification is the **substrate-graph analog of standard Morse theory** applied to the substrate-action functional. It is form-IDENTIFIED at substrate-graph level via SC-4.9 + standard differential geometry of critical points.

§3.2 Kernel-induced curvature

V1 + V5 kernels enter the Hessian via the kernel-derivative terms in $\mathcal{L}_{\text{sub}}[\Psi, \nabla_{V_1} \Psi, \nabla_{V_5} \Psi]$:

$$\mathcal{H}_{\text{kernel}}(\ell, \ell') = \sum_{K \in \{V_1, V_5\}} G_K \cdot K(\ell - \ell')$$

where G_K are kernel-derivative coupling coefficients. Per Memo_ED_RouteA_A2_Construct §2.2, the kernel terms have long-wavelength expansion $\tilde{K}(q) \rightarrow G_K \ell_K^2 q^2$ for $q \ll 1/\ell_K$, giving IR-limit dispersion $\lambda(q) = U''(\Psi^{\text{const}}) + \beta q^2$ where $\beta = \sum_K G_K \ell_K^2$.

The **kernel-induced curvature** is the substrate-graph analog of the standard QFT field-curvature term; its sign determines whether V1/V5 propagation around the saddle is stable (positive curvature; oscillatory) or unstable (negative curvature; growth/collapse).

§3.3 Saddle identification — compression vs expansion directions

At a generic non-degenerate substrate-action saddle, the Hessian eigenvalue spectrum has mixed signature. The orthogonal directions partition into:

- **Compression-dominant axes** ($\lambda_i < 0$): substrate-action increases under contraction along $\vec{u}_i \rightarrow$ local dynamics favor *collapse* in $\vec{u}_i$ direction
- **Expansion-dominant axes** ($\lambda_i > 0$): substrate-action decreases under contraction along $\vec{u}_i \rightarrow$ local dynamics favor *expansion* in $\vec{u}_i$ direction

The number of compression-dominant axes vs expansion-dominant axes defines the *Morse index* of the saddle. The three universal regime classes of §1 correspond to Morse-index extremes (d_{sub} or 0, all-compression or all-expansion) and the intermediate proper-saddle range ($0 < \text{index} < d_{\text{sub}}$).

This is form-IDENTIFIED via SC-4.9 + standard Morse theory; no substrate-research-frontier work required.

§3.4 Local dynamics from saddle geometry

The substrate-side dynamical response at a saddle locus is generated by V1 retarded propagation (Paper_089) of the Hessian-signature-pattern reconfiguration at substrate-c (Paper_012 P-RB-1), with V5 cross-chain coupling (Paper_090) supplying long-range correlation content. At leading order in $\varepsilon = \ell_{\text{ED}}/R_{\text{cg}}$, DCGT (Paper_073) coarse-grains this substrate-graph response to continuum dynamical behavior.

The **characteristic length scale** of the dynamical response is set by the IR Hessian eigenvalue:

$$\ell_{\text{saddle}} \sim 1/\sqrt{|\lambda_{\text{IR}}|}$$

For uniform- Ψ saturation (Cos_01 / Cos_05 regimes), $\lambda_{\text{IR}} \rightarrow 0$ at the kernel-set scale, and $\ell_{\text{saddle}} \rightarrow \ell_{V_5} = c\tau_{V_5}$ — i.e., the V5 finite-memory length, which is identified with $R_H = c/H_0$ via Route A4 (Memo_ED_RouteA_A4_Construct; per multi-route convergence audit option (ii) at substrate-parameter-INHERITED level). For non-saturation regimes (Dyn_02 RDE/MDE territory), ℓ_{saddle} scales with the local Hessian-eigenvalue magnitudes per the kernel-induced curvature term.

Substrate-c bound preserved throughout; exponential or other scaling emerges at continuum-side via DCGT translation — same mechanism standard cosmology uses for de Sitter expansion.

§3.5 Relation to Cos_01 + Cos_05 saturation regimes

The all-expansion ($\lambda_i > 0$ all i) regime corresponds to **substrate-graph saturation** — uniform Ψ with V1 + V5 + individuation reactivated (per Paper_087 P12 + Paper_ED_CCC §3.7). Under uniform Ψ , kernel derivatives $\nabla_K \Psi$ vanish (Memo_ED_ChainClass_C3_Construct §3.1); the saddle Hessian degenerates to all-eigenvalues-equal at the saturation extremum, with the all-expansion sign-structure. Q1A + Q2A composition (Memo_ED_Q1Q2_JointClosure_Construct) gives vacuum-energy form continuum stress-energy with $w = -1$; Friedmann recovery gives de Sitter / exponential expansion.

This is the foundational dynamical content of Cos_01 (inflation) and Cos_05 (dark energy): both are the all-expansion saddle-dynamical regime at different temporal locations in the cosmological timeline (post-BBB ignition vs late-LDE asymptote). Dyn_01’s saddle-as-universal-generator framework subsumes both as instances of the same dynamical regime.

§3.6 Relation to Dyn_02 (horizon motion)

Mixed-signature (S2) saddles correspond to **horizon-class dynamics**: substrate loci with some compression-dominant + some expansion-dominant axes. The boundary between Hessian-signature-classes across the substrate-graph is a *horizon* — the substrate-graph analog of the boundary between causally-connected and causally-disconnected substrate content (Paper_ED_GW_00 §3.1: “the horizons are the dynamics”). Horizon motion = evolution of this Hessian-signature boundary under V1 + V5 + DCGT dynamics.

This is the foundational dynamical content of Dyn_02 (horizon-motion law): Dyn_02’s non-saturation regime identifications (RDE $w = 1/3$, MDE $w = 0$) inherit the mixed-signature saddle-dynamical framework; the RDE/MDE chain-class identifications (load-bearing OPEN per Q1/Q2 inheritance) are inherited from Dyn_02 unchanged.

§3.7 Relation to Dyn_03 (radiation law) and GW_00

Time-varying Ψ at a saddle locus produces propagating Hessian-signature reconfiguration carried by V1 retarded propagation. This is the substrate-graph mechanism for **radiation** (Dyn_03 — Larmor + GW quadrupole forms) and for **gravitational-wave propagation** (GW_00 — saddle-Hessian-signature reconfiguration as the propagating “wave”). Both are instances of saddle-Hessian-signature dynamics in the time-varying regime.

For Dyn_03 source-class identification (accelerated chain = EM; time-varying multipole = GW), the saddle-dynamical framework supplies the *structural* identification; the *specific* identifications (Q1/Q2 inheritance OPENs in Dyn_03) remain OPEN at the same level. For GW_00 row 12 (quadrupole source-amplitude formula), the closure via NoetherFlux substrate-graph chain inherits Dyn_01’s saddle-dynamical framework as foundational substrate content.

§4 Audit Table

#	Step	Label	Notes
1	P02, P03, P09, P10, P11, P12, P13	P	Paper_087. P03 + P13 supply Noether translation invariance; P12 supplies Hessian-signature relevance.
2	Upstream inheritance block	I	Paper_012; Paper_027; Paper_073 (DCGT A→regime); Paper_089; Paper_090; Paper_093; Paper_ED_SC_4_9; Paper_ED_Cos_01 + Cos_05; Paper_ED_Dyn_02 + Dyn_03; Paper_ED_GW_00; Up-date_ED_SC_4x_Arc_M3.
3	Standard differential geometry / Morse-theory machinery	I	Textbook inheritance; no substrate-graph derivation claimed.

#	Step	Label	Notes
4	Substrate-action Hessian eigenvalue decomposition at each substrate locus	D-via-I	§3.1. Composition of Paper_ED_SC_4_9 Hessian framework + standard Morse-theory.
5	Hessian-signature partition into compression-dominant ($\lambda < 0$) and expansion-dominant ($\lambda > 0$) axes	D-via-I	§3.1, §3.3. Inherits SC-4.9 S1/S2/S3 classification.
6	Kernel-induced curvature from V1 + V5 long-wavelength expansion	D-via-I	§3.2. Composition of Paper_089 + Paper_090 + Memo_ED_RouteA_A2_Construct §2.2 dispersion analysis.
7	Three universal regime classes (all-compression S1; all-expansion S3; mixed S2)	D-via-I	§3.3 + §1 table. Composition of rows 4, 5, 6 with SC-4.9 classification.
8	All-expansion regime saturation regime (Cos_01 + Cos_05)	D-via-I	§3.5. Inherits Q1A + Q2A + C3 substrate-graph chain (Memo_ED_Q1Q2_JointClosure_Co + Memo_ED_ChainClass_C3_Construct
9	Mixed-signature regime horizon dynamics (Dyn_02)	D-via-I	§3.6. Composition of row 5 + Paper_ED_GW_00 §3.1 (“horizons are the dynamics”) + Paper_ED_Dyn_02.
10	Time-varying saddle-Hessian reconfiguration radiation + GW propagation (Dyn_03 + GW_00)	D-via-I	§3.7. Composition of row 4 + Paper_ED_Dyn_03 + Paper_ED_GW_00.
11	Substrate-c bound preserved throughout	D-via-I	§3.4. Paper_012 P-RB-1 + Paper_089 + Paper_093.

#	Step	Label	Notes
12	DCGT translation of substrate-graph saddle dynamics to continuum dynamical behavior	D-via-I	§3.4. Paper_073 hydrodynamic-window coarse-graining.
13	Characteristic length $\ell_{\text{saddle}} \sim 1/\sqrt{ \lambda_{\text{IR}} }$; at saturation $\rightarrow \ell_{V_5} \sim R_H$	D-via-I	§3.4. IR Hessian-eigenvalue scaling + Route A4 substrate-parameter-INHERITED identification.
14	ℓ_{saddle} at substrate-parameter-INHERITED level via Route A4 (Paper_027-analog)	I	Sub-type per label-class note: <i>substrate-parameter-INHERITED</i> . §3.4. Per Memo_ED_RouteA_MultiRouteCon option (ii); $\Theta_{\text{ED}}^{\text{Planck-units}} \approx 10^{-122}$ + $\tau_{V_5}/\tau_P \approx 10^{61}$ at primitive level.
15	Local Hessian-signature + Ψ -gradient structure per substrate state	I	Sub-type per label-class note: <i>IC-INHERITED</i> . §3.4. Framework-neutral per Memo_ED_CCC_Realizability_Sub (cyclic-CCC or R3 single-aeon).
16	Verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A $\rightarrow$ regime	A$\rightarrow$position	Per Paper_095 §3.3 line 77. Five-anchor verdict-sync: status / abstract / §1 / this row / §6 all M3.

Audit summary: zero pure-D rows; **nine substantive D-via-I rows** (4–13); two I rows (2, 3); one substrate-parameter-INHERITED row (14); one IC-INHERITED row (15); one P row (1); one A $\rightarrow$ position verdict row (16). **No load-bearing OPEN items** — all open content is inherited from upstream papers’ OPEN status (Dyn_02/Dyn_03 Route C1/C4; SC-4.9 substrate-FORCED exhaustiveness) and is tightening-work-only at Dyn_01’s foundational level. No new substrate primitives; no paper-specific postulates.

§5 Falsification Criteria

- **F1 (sharpest):** Observation of dynamical behavior at a substrate-graph saddle locus inconsistent with the Hessian-signature partition — e.g., compression-dominant collapse behavior at an all-expansion S3-saddle, or expansion-dominant behavior at an all-compression S1-saddle. Refutes the saddle-as-foundational-generator identification at §1 + §3.3. Currently consistent: cosmological-scale all-expansion (Cos_01 + Cos_05) and BH-horizon mixed-signature (GW_00) behavior match the predicted regimes.
- **F2:** Substrate-graph proof that $\ell_{\text{saddle}} \sim \ell_{V_5}$ at saturation — i.e., the IR Hessian-eigenvalue scaling does not yield the Route A4 substrate-parameter scale at the saturation regime. Refutes §3.4 + audit row 13.
- **F3:** Failure of saddle-Hessian-signature framework to reproduce Paper_ED_GW_00 quadrupole-formula scaling ($P_{\text{GW}} \propto |\ddot{Q}_{ij}|^2$) for time-varying mixed-signature saddle configurations. Refutes §3.7 + audit row 10.
- **F4:** Failure of saddle-dynamical framework to reproduce Paper_ED_Dyn_02 non-saturation regime horizon-motion phenomenology (Friedmann $a(t)$ scalings per regime). Refutes §3.6 + audit row 9.

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

ED dynamics is the substrate-graph saddle Hessian-signature dynamics of $S_{\text{sub}}[\Psi]$, carried by V1 + V5 kernel propagation under DCGT translation. The Hessian-signature partition (Paper_ED_SC_4_9 S1/S2/S3 classification + standard Morse theory) gives three universal dynamical regime classes: all-compression (gravitational collapse), all-expansion (inflationary / dark-energy saturation, per Cos_01 + Cos_05), mixed-signature (horizon-class dynamics per Dyn_02 + propagating signature-reconfiguration per GW_00). Characteristic length ℓ_{saddle} sets the dynamical scale; at saturation, $\ell_{\text{saddle}} \rightarrow R_H = c/H_0$ at substrate-parameter-INHERITED level via Route A4 (Paper_027-analog) per the 2026-05-17 Route A audit cascade closure. Substrate-c bound preserved throughout.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window per Paper_095. Nine D-via-I rows; one substrate-parameter-INHERITED row (Route A4); one IC-INHERITED row (framework-neutral per Memo_ED_CCC_Realizability_SubConstruct); zero pure-D rows; no paper-specific postulates; no new substrate primitives; **no load-bearing OPEN items**. Upstream OPEN content (Dyn_02/Dyn_03 chain-class identifications; SC-4.9 substrate-FORCED exhaustiveness; Route A tightening RA-OPENs) is inherited at the upstream papers' status — Dyn_01 does not aggravate or depend on those OPENs for its M3 verdict.

Cross-arc consequences. Upstream, Cos_01 + Cos_05 (M3) become reinterpretable as the all-expansion saddle-dynamical regime; Dyn_02 + Dyn_03 (M2) as mixed-signature + time-varying

saddle-dynamical regimes; GW_00 (M3) as time-varying mixed-signature reconfiguration; the ED-SC 4.x arc-wide M3 picture as six SCBU projections of saddle dynamics at different substrate-parameter axes. Downstream, Dyn_01 *enables* drafts of Paper_ED_Dyn_04 (Gravitational Collapse) by supplying the S1-class regime template, Paper_ED_Dyn_05 (Inspirational Dynamics) by supplying mixed-signature + time-varying framework jointly with Dyn_03, and Paper_ED_GW_01 (BH Ringdown Spectroscopy) by supplying the mixed-signature template jointly with GW_00. Whether each downstream paper closes at M3 or M2 depends on its own substrate-graph chain (chain-class identification + Q1A/Q2A construction-uniqueness + Route A4 inheritance) and is not forecast here.

End Paper_ED_Dyn_01.